Hints: Linear algebra – 8. Block matrices and determinants of special shape

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Schur formula: $\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} E & 0 \\ CA^{-1} & E \end{pmatrix} \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}$. Compute the determinant of $\begin{pmatrix} E & -B \\ A & E \end{pmatrix}$ using either diagonal block as the pivot: you get $\det(E+BA)$ and $\det(E+AB)$.

Hint 2. Commuting C, D: $\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} D & 0 \\ -C & E \end{pmatrix} = \begin{pmatrix} AD - BC & B \\ 0 & D \end{pmatrix}$. Cancel $\det D$ if D is invertible; otherwise replace D by $D + \varepsilon E$, which still commutes with C, and let $\varepsilon \to 0$.

Hint 3. Counterexample (2×2 blocks): $A = E$, $B = E_{21}$, $C = E_{12}$, $D = E_{11}$. Here $\det M = \det(D - CB) = 0$ but $\det(AD - BC) = -1$.

Problem 2

Hint 1. Note the clash of notation: here the block E of M^{-1} is not the identity. Write I for the identity. Multiply M by the block-triangular matrix $\begin{pmatrix} I & F \\ 0 & H \end{pmatrix}$, built from the right block column of M^{-1}.

Hint 2. Since $MM^{-1} = I$, we have $AF + BH = 0$ and $CF + DH = I$. So the product is $\begin{pmatrix} A & 0 \\ C & I \end{pmatrix}$. Take determinants.

Problem 3

Hint 1. Let $X = ABCD$. Then $X = A(BCD) = AA^T$ is symmetric positive semidefinite. Find a polynomial identity for X.

Hint 2. Transpose $A^T = BCD$ and substitute the other three relations: $A = D^T C^T B^T = (ABC)(DAB)(CDA)$, i.e. $X^2 A = A$.

Hint 3. Hence $X^3 = X^2 AA^T = AA^T = X$. A positive semidefinite matrix with $X^3 = X$ has eigenvalues in $\{0, 1\}$ and is diagonalisable, so $X^2 = X$.

Problem 4

Hint 1. Since n is odd, $i \mapsto 2i$ is a bijection of $\mathbb{Z}/n$. Relabel the indices by it: A becomes the circulant $2E + P + P^{-1}$, where P is the cyclic shift.

Hint 2. $2E + P + P^{-1} = P^{-1}(E + P)^2$, and $\det(E + P) = \prod_k (1 + \varepsilon^k) = 1 - (-1)^n = 2$. Answer: $\det A = 4$.

Problem 5

Hint 1. C_n is the circulant with symbol $f(x) = 1 + x + x^2 = (x - \omega)(x - \omega^2)$, $\omega = e^{2\pi i/3}$. So $\det C_n = \prod_k f(\varepsilon^k)$. If $3 \mid n$, some $\varepsilon^k = \omega$.

Hint 2. $\prod_k(\varepsilon^k - z) = (-1)^n(z^n - 1)$ because $\prod_k(z - \varepsilon^k) = z^n - 1$. Hence $\det C_n = (\omega^n - 1)(\omega^{2n} - 1) = |\omega^n - 1|^2$, which is 3 when $3 \nmid n$.